

JavaScript Data Types

What is a Data Type?

A **Data Type** tells JavaScript what kind of value a variable stores.

```
let name = "Rokon"; // String
let age = 23; // Number
```

Types of Data

JavaScript has **2 main types** of data:

1. Primitive Data Types
2. Non-Primitive (Reference) Data Types

1. Primitive Data Types

Primitive data types store a **single value** and are **immutable**.

There are **7 primitive data types**:

Data Type	Description	Example
String	Text	"Hello"
Number	Integer or Decimal	10 , 3.14
Boolean	true or false	true
Undefined	Variable declared but not assigned	undefined
Null	Intentionally empty value	null
BigInt	Very large integer	123456789n
Symbol	Unique identifier	Symbol("id")

Examples

```
let name = "Rokon";
let age = 23;
let isStudent = true;
let city;
let car = null;
let bigNumber = 12345678901234567890n;
let id = Symbol("userId");
```

2. Non-Primitive (Reference) Data Types

Non-Primitive data types can store **multiple values** and are stored **by reference**.

Object

```
const student = {
  name: "Rokon",
  age: 23,
};

console.log(student.name);
```

Array

```
const fruits = ["Apple", "Mango", "Orange"];

console.log(fruits[0]);
```

Function

```
function greet() {  
  console.log("Hello!");  
}  
  
greet();
```

Primitive vs Non-Primitive

Feature	Primitive	Non-Primitive
Stores	Single Value	Multiple Values
Mutable	No	Yes
Copied By	Value	Reference
Examples	String, Number, Boolean	Object, Array, Function

Primitive Example

```
let a = 10;  
let b = a;  
  
b = 20;  
  
console.log(a); // 10  
console.log(b); // 20
```

Explanation: `b` gets a copy of `a`, so changing `b` does not affect `a`.

Non-Primitive Example

```
let person1 = {  
  name: "Rokon",  
};  
  
let person2 = person1;
```

```
person2.name = "Rahim";

console.log(person1.name); // Rahim
console.log(person2.name); // Rahim
```

Explanation: Both variables point to the same object, so changing one also changes the other.

Easy Way to Remember

- **Primitive** = Each variable has its **own copy**.
- **Non-Primitive** = Multiple variables can point to the **same object**.

Interview Question

Q: What is the difference between Primitive and Non-Primitive Data Types?

Answer:

Primitive data types store a single value and are copied **by value**, while Non-Primitive data types store multiple values and are copied **by reference**.